

CSS Tables

Complete Guide with Code Examples & Practical Usage

Introduction

CSS tables allow you to present data in organized, structured formats. By styling tables effectively with CSS, you can make data more readable, visually appealing, and easier to understand. This comprehensive guide covers table styling techniques, responsive tables, and best practices for professional data presentation.

Table CSS Properties Reference

Property	Description	Example Values
border-collapse	Merge or separate table borders	collapse, separate
border-spacing	Space between cells	0, 5px, 10px
table-layout	How table calculates width	auto, fixed
width	Table width	100%, 800px, 90%
text-align	Align cell content	left, center, right
vertical-align	Vertical cell alignment	top, middle, bottom

Example 1: Basic Table Styling

Clean, professional table with minimal styling.

CSS Code:

```
table { width: 100%; border-collapse: collapse; margin: 20px 0; } th, td { padding: 12px; text-align: left; border-bottom: 1px solid #ddd; } th { background-color: #3f51b5; color: white; font-weight: bold; }
```

HTML Usage:

```
<table> <thead> <tr> <th>Header 1</th> <th>Header 2</th> </tr> </thead> <tbody> <tr> <td>Data 1</td> <td>Data 2</td> </tr> </tbody> </table>
```

Live Demo:

Name	Department	Salary
John Smith	Engineering	\$75,000
Sarah Johnson	Marketing	\$65,000
Mike Chen	Sales	\$70,000

Example 2: Striped Row Table

Alternate row colors for improved readability.

CSS Code:

```
/* Striped rows */ tbody tr:nth-child(odd) { background-color: white; } tbody tr:nth-child(even) { background-color: #f0f4ff; } /* Hover effect */ tbody tr:hover { background-color: #e8eaf6; }
```

HTML Usage:

```
<table> <tbody> <tr><td>Row 1</td></tr> <tr><td>Row 2</td></tr> </tbody> </table>
```

Live Demo:

Product	Quantity	Price
Laptop	5	\$800
Monitor	8	\$300
Keyboard	12	\$80
Mouse	15	\$25

Example 3: Bordered Table

Table with visible borders on all cells.

CSS Code:

```
table { border: 2px solid #3f51b5; border-collapse: collapse; } th, td { border: 1px solid #3f51b5; padding: 12px; } th { background-color: #3f51b5; color: white; }
```

HTML Usage:

```
<table style="border: 2px solid #3f51b5;"> ...table content... </table>
```

Live Demo:

Q1	Q2	Q3	Q4
\$100K	\$120K	\$110K	\$150K
\$95K	\$105K	\$115K	\$125K

Example 4: Interactive Hover Table

Highlight rows on hover for interactivity.

CSS Code:

```
/* Hover highlight */ tbody tr:hover { background-color: #e8eaf6; cursor: pointer; box-shadow: 0 2px 4px rgba(0,0,0,0.1); } /* Smooth transition */ tbody tr { transition: all 0.3s ease; }
```

HTML Usage:

```
<table style="cursor: pointer;"> ...table content... </table>
```

Live Demo (Hover over rows):

User	Status	Date
Alice	Active	2026-05-04
Bob	Inactive	2026-05-03
Charlie	Active	2026-05-02

Example 5: Compact Table

Reduced padding for dense data display.

CSS Code:

```
/* Compact styling */ th, td { padding: 6px; font-size: 14px; } /* Dense borders */ table { border-collapse: collapse; } th { background: #3f51b5; color: white; }
```

HTML Usage:

```
<table class="compact"> ...table content... </table>
```

Live Demo:

ID	Item	Qty
001	Apple	10
002	Banana	15
003	Orange	12
004	Grape	8

Example 6: Fixed Width Columns

Control column widths precisely.

CSS Code:

```
/* Fixed table layout */ table { width: 100%; table-layout: fixed; } /* Column specific widths */ th:nth-child(1), td:nth-child(1) { width: 30%; } th:nth-child(2), td:nth-child(2) { width: 40%; } th:nth-child(3), td:nth-child(3) { width: 30%; } /* Text wrapping */ td { word-wrap: break-word; }
```

HTML Usage:

```
<table style="table-layout: fixed;"> ...table content... </table>
```

Live Demo:

Name	Description	Price
Product A	High quality item with great features	\$50
Product B	Affordable option for budget conscious	\$30

Example 7: Responsive Table

Tables that adapt to screen size.

CSS Code:

```
/* Desktop table */ table { width: 100%; border-collapse: collapse; } /* Responsive on mobile */ @media (max-width: 768px) { table, thead, tbody, th, td, tr { display: block; } th { position: absolute; top: -999px; left: -999px; } td { position: relative; padding-left: 50%; } td:before { content: attr(data-label); position: absolute; left: 6px; font-weight: bold; } }
```

HTML Usage:

```
<table> <tr> <td data-label="Name">John</td> <td data-label="Email">john@example.com</td> </tr> </table>
```

Note:

Responsive tables adapt layout on smaller screens. Test by resizing your browser or viewing on mobile devices.

Example 8: Table Caption & Accessibility

Add captions and semantic markup for better accessibility.

CSS Code:

```
/* Caption styling */ caption { caption-side: top; padding: 10px; font-weight: bold; color: #3f51b5; text-align: left; } /* Semantic markup */ thead { background: #3f51b5; color: white; } tbody { background: #f0f4ff; font-weight: bold; }
```

HTML Usage:

```
<table> <caption>Sales Data Q1-Q4 2025</caption> <thead>...</thead> <tbody>...</tbody> <tfoot>...</tfoot> </table>
```

Live Demo:

Annual Revenue Report		
Year	Revenue	Growth
2023	\$1.2M	10%
2024	\$1.5M	25%
2025	\$1.9M	27%

Best Practices

1. Use Semantic HTML

```
/* Use thead, tbody, tfoot */ <table> <thead>...headers...</thead> <tbody>...data...</tbody> <tfoot>...totals...</tfoot> </table>
```

2. Make Tables Accessible

```
/* Add scope and headers */ <th scope="col">Column Header</th> <th scope="row">Row Headers</th> /* Add caption */ <caption>Descriptive title</caption>
```

3. Optimize for Mobile

```
/* Responsive tables */ @media (max-width: 768px) { table { font-size: 12px; } th, td { padding: 8px; } /* Or convert to card layout */ /* Depends on data complexity */ }
```

4. Proper Styling Hierarchy

```
/* Clear visual hierarchy */ th { background: #3f51b5; color: white; font-weight: bold; } td { padding: 12px; } tbody tr:nth-child(even) { background: #f9f9f9; } tbody tr:hover { background: #e8eaf6; }
```

5. Keep Tables Simple

```
/* Avoid complex nested tables */ /* Don't merge too many cells */ /* Keep column count reasonable (5-8 optimal) */ /* Break large tables into sections */
```

💡 **Tip:** Use `border-collapse: collapse` to merge table borders into a single line, creating a cleaner appearance.

Common Issues & Solutions

Issue: Uneven Column Widths

Problem:

```
table { width: 100%; } /* Column widths vary based on content */
```

Solution:

```
table { width: 100%; table-layout: fixed; } /* Distribute width evenly */
```

Issue: Text Doesn't Wrap in Cells

Problem:

```
td { /* Long text overflows */ }
```

Solution:

```
td { word-wrap: break-word; overflow-wrap: break-word; max-width: 200px; }
```

Issue: Table Too Wide for Mobile

Problem:

```
table { width: 100%; } /* Horizontal scroll required on mobile */
```

Solution:

```
/* Use responsive CSS or wrap in scrollable container */ @media (max-width: 768px) { .table-wrapper { overflow-x: auto; } }
```

Summary

- Use `border-collapse: collapse` for cleaner tables
- Apply `nth-child(even)` for striped rows
- Use `tbody tr:hover` for interactive feedback
- Style `thead`, `tbody`, `tfoot` semantically
- Use `table-layout: fixed` for consistent widths
- Add `caption` for table titles
- Make tables `responsive` for mobile
- Maintain proper `padding` and spacing
- Ensure `color contrast` for readability
- Include scope attributes for accessibility
- Test tables across devices and browsers